

Quick Reference - Universal Wells System

File Locations

File	Location	Purpose
wells_config.php	(root)	Central config for all wells
config.php	Each well dir	Loads well-specific settings
api.php	Each well dir	API functions (shared)
index.php	Each well dir	Main well page
monthly.php	Each well dir	Monthly data view
fetch_data.php	Each well dir	AJAX endpoint

Adding a New Well (3 Steps)

1. Edit wells_config.php

```
php
'newwell' => [
    'well_id' => 'newwell',
    'location_id' => 'GET_FROM_HYDROVU',
    'common_name' => 'New Well',
    'full_name' => 'New Well Full Name',
    'water_level_baseline' => 800.00,
    'water_level_warning' => 830.00,
    'description' => 'Description here',
    'param_names' => ['depth' => 'Groundwater Level Elevation', 'temperature' => 'Temperature'],
    'param_units' => ['depth' => 'ft', 'temperature' => '°F']
]
```

2. Create Directory & Copy Files

```
bash
mkdir newwell
cp hp/*.php newwell/
```

3. Test

Visit: <yoursite.com/newwell/>

Key Features

Universal Configuration

- ✔ One config file for all wells
- ✔ Auto-detection of well from URL
- ✔ Easy to add/remove wells

Loading Improvements

- ✔ Skeleton loader (no blank page)
- ✔ Smooth CSS transitions
- ✔ Better perceived performance

Caching

- ✔ 15-minute data cache
- ✔ 24-hour metadata cache
- ✔ Reduces API load

Debug Mode

Add `?debug=1` to any URL:

```
yoursite.com/hp/?debug=1
yoursite.com/hickman1/?debug=1
```

Shows:

- Current well ID
- Location ID
- API timing
- Cache status
- Data processing info

URL Structure

```
yoursite.com/      → Well directory
yoursite.com/hp/   → Horse Park main
yoursite.com/hp/monthly.php → Horse Park monthly
yoursite.com/hickman1/ → Hickman1 main
yoursite.com/hickman1/monthly.php → Hickman1 monthly
```

⚙️ Configuration Values Explained

Field	Example	Description
<code>well_id</code>	<code>hp</code>	URL identifier (lowercase, no spaces)
<code>location_id</code>	<code>{5515870...}</code>	HydroVu API location ID (16 digits)
<code>common_name</code>	<code>'Horse Park'</code>	Short display name
<code>full_name</code>	<code>'Kentucky Horse Park...'</code>	Full page title
<code>water_level_baseline</code>	<code>804.25</code>	Elevation baseline (feet)
<code>water_level_warning</code>	<code>838</code>	Alert threshold (feet)

🚨 Common Issues

"Could not determine which well"

- ✓ Check directory name matches `well_id`
- ✓ Verify `wells_config.php` exists in root
- ✓ Check well is in config array

Wrong Data Showing

- ✓ Verify `location_id` is correct
- ✓ Clear cache (delete cache files)
- ✓ Check debug mode output

Slow Loading

- ✓ Check API response time (debug mode)
- ✓ Verify cache is working
- ✓ Skeleton should show immediately

💡 Pro Tips

1. **Test with Horse Park first** - It's already configured
2. **Use debug mode** - Shows exactly what's happening
3. **Copy files exactly** - Don't modify unless needed
4. **Keep backups** - Before making changes
5. **Check location IDs** - Most common error source

📍 Getting Location ID

From HydroVu API or dashboard:

```
php
// Make request to: /locations/list
// Response includes all your locations with IDs
```

🛠️ Customization Points

Per-Well Settings

- Baseline elevation
- Warning thresholds
- Parameter names
- Unit displays

Global Settings

- API credentials
- Cache duration
- Time ranges
- Debug mode

✅ Verification Checklist

Before going live with a new well:

- ☐ Added to `wells_config.php`
- ☐ Directory created with correct name
- ☐ All PHP files copied
- ☐ Location ID verified
- ☐ Baseline/warning set correctly
- ☐ Tested with `?debug=1`
- ☐ Both index and monthly work
- ☐ Data displays correctly
- ☐ Charts render properly
- ☐ Loading skeleton works

📊 Performance Metrics

With caching enabled:

- First load: ~2-3 seconds (API call)
- Cached load: <500ms (no API call)

- Skeleton: <100ms (instant)
- Cache refresh: Every 15 minutes

Security Notes

- Keep API credentials secure
- Restrict (`wells_config.php`) permissions
- Remove (`?debug=1`) in production
- Validate all inputs (already handled)

File Sizes

Approximate sizes after setup:

- (`wells_config.php`): ~2-5 KB
- (`config.php`): ~2 KB
- (`api.php`): ~15-20 KB
- (`index.php`): ~10-15 KB
- (`monthly.php`): ~8-12 KB

Learning Path

1. Understand current HP setup ✓
2. Study `wells_config.php` ✓
3. Add Hickman1 to config
4. Create Hickman1 directory
5. Copy files and test
6. Add third well
7. Master the system!